CMSC 420: Bloom Filters

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1 Problems **2**

1 Problems

1. Suppose we define a Bloom filter using $S = \mathcal{Z}_{12}$, with B indexed by $\mathcal{Z}_{11}$, and with two hash functions:

$$h_1(x) = 2x \mod 11$$
$$h_2(x) = 3x \mod 11$$

(a) Suppose you insert the keys $x = 1, 6, 9$. Show the state of the Bloom filter after each insertion.

Solution:

Insert	$h_1(x)$	$h_2(x)$	B
$x = 1$	2	3	00110000000
$x = 3$	1	7	01110001000
$x = 9$	7	5	01110101000

(b) Identify all false positives.

Solution:

Let's just check all the other keys:

Key	$h_1(x)$	$h_2(x)$	FP?
$x = 0$	0	0	No
$x = 2$	4	6	No
$x = 4$	8	1	No
$x = 5$	10	4	No
$x = 6$	1	7	No
$x = 7$	3	10	No
$x = 8$	5	2	Yes
$x = 10$	9	8	No
x $x = 11$	0	0	No

Yes, we get a false positives for $x = 8$.

2. Suppose we define a very small Bloom filter with $m = 3$ and $k = 2$.

   (a) We insert $n = 1$ keys. Write down all the possible resulting bit arrays and the probability of each.

   Solution:
   We have:
   `100`: $p = 1/9$
   `010`: $p = 1/9$
   `001`: $p = 1/9$
   `110`: $p = 2/9$
   `101`: $p = 2/9$
   `011`: $p = 2/9$

   (b) We insert $n = 2$ keys. What is the probability that the resulting bit array is `100`?

   Solution:
   When we insert $n = 2$ keys with $k = 2$ hash functions we are essentially picking four indices between 0 and 2 with repeats permitted and setting those positions to `1`.
   To obtain `100` we must choose index 0 four times in order.
   The number of ways to choose four 0s in order is 1 and it has probability 1/81.

   (c) We insert $n = 2$ keys. What is the probability that the resulting bit array is `110`?

   Solution:
   When we insert $n = 2$ keys with $k = 2$ hash functions we are essentially picking four indices between 0 and 2 with repeats permitted and setting those positions to `1`.
   To obtain `110` we must choose between indices 0 and 1 four times in order but the result must include at least one 0 and one 1.
   The number of ways to choose four 0s and 1s in order is $2^4 = 16$ but we have to exclude 0,0,0,0 and 1,1,1,1 so there are 14 ways.
   Each has probability 1/81 and they are disjoint so the result is 14/81.

3. Suppose we define a Bloom filter with $m \geq 2$ and $k = 2$. We insert $n = 2$ keys.

   (a) What is the probability (in terms of m) that the resulting bit array has `1`s in exactly one index and `0`s elsewhere?

   Solution:
   When we insert $n = 2$ keys with $k = 2$ hash functions we are essentially picking four indices between 0 and $m-1$ with repeats permitted and setting those entries to `1`. There are m^4 ways to do this, each with equal probability of $1/m^4$.
   Consider a specific index x. To obtain `1`s in only that index we must choose from the set of indices $\{x\}$ four times in a row.
   The number of ways to choose from the set of indices $\{x\}$ four times in a row is exactly 1.
   However there are $C(m, 1) = m$ ways of choosing a specific index x so there are a total of $1m = m$ ways.
   Since (as mentioned in the opening) each has probability $1/m^4$ and since they are disjoint the result is $m/m^4 = 1/m^3$.

   (b) What is the probability (in terms of m) that the resulting bit array has `1`s in exactly two indices and `0`s elsewhere?

   Solution:
   Consider two specific distinct indices x and y. To obtain `1`s in only those two indices we must choose from the set of indices $\{x, y\}$ four times in a row but the result must include at least one of each.
   The number of ways to choose from the set of indices $\{x, y\}$ four times in a row is $2^4 = 16$ but we have to exclude the two choices x, x, x, x and y, y, y, y so there are 14 ways.
   However there are $C(m, 2) = m(m-1)/2$ ways of choosing two specific distinct indices x and y so there are a total of $14m(m-1)/2 = 7m(m-1)$ ways.
   Since (as mentioned in the opening to part (a)) each has probability $1/m^4$ and since they are disjoint the result is $7m(m-1)/m^4 = 7(m-1)/m^3$.

4. Using the inaccurate (but not terrible) bloom filter false positive expression:

$$p = \left(1 - \left(1 - \frac{1}{m}\right)^{kn}\right)^k$$

(a) Suppose we wish to store $n = 70$ items with $k = 3$ hash functions and a bit array of size $m = 50$. What would the corresponding false positive probability be? Give an exact answer (unsimplified) and an approximation to four decimal digits.

Solution:
It would be:

$$p = \left(1 - \left(1 - \frac{1}{50}\right)^{(3)(70)}\right)^3 \approx 0.9575$$

(b) Suppose we wish to store $n = 100$ items and we want a false positive rate no higher than 2%. Find a restriction on m (as a function of k) which will guarantee this.

Solution:
We want $p \leq 0.02$ and thus:

$$\begin{aligned}
\left(1 - \left(1 - \frac{1}{m}\right)^{100k}\right)^k &\leq 0.02 \\
1 - \left(1 - \frac{1}{m}\right)^{100k} &\leq \sqrt[k]{0.02} \\
\left(1 - \frac{1}{m}\right)^{100k} &\geq 1 - \sqrt[k]{0.02} \\
1 - \frac{1}{m} &\geq \sqrt[100k]{1 - \sqrt[k]{0.02}} \\
\frac{1}{m} &\leq 1 - \sqrt[100k]{1 - \sqrt[k]{0.02}} \\
m &\geq \frac{1}{1 - \sqrt[100k]{1 - \sqrt[k]{0.02}}}
\end{aligned}$$

5. Suppose the state of a particular count array for a Counting Bloom Filter after n insertions with k hash functions is:

```
B = [3,1,2,1,0,2,2,0,0]
```

Suppose you don't know k but you know that n is a single-digit number. Moreover suppose that for any given key there are at least two hash functions which produce different outputs. Prove that there are at least six distinct hash functions.

Solution:
Each insert increases the total by k so the overall total will be nk.
In this case the sum is 11 so $nk = 11$. The only solutions are $n = 11, k = 1$ or $n = 1, k = 11$. However we know n is a single-digit number so we have $n = 1, k = 11$.
Since only 1 key x_0 was inserted at least six of those hash functions must be distinct or else we would have five or fewer outputs among $h_1(x_0), ..., h_{11}(x_0)$ and then we could not have six non-zero entries in our count array.

6. Suppose the state of a particular count array for a Counting Bloom Filter after n insertions with k hash functions is:

```
B = [2,1,2,2,1,5,1,0,0]
```

Suppose you know both k and n are single-digit numbers and you know that for any given key no more than two of the hash functions will give identical results.

What are n and k? Justify.

Solution:
Each insert increases the total by k so the overall total will be nk.
In this case the overall total is 14 so $nk = 14$. Because of the single-digit restriction we cannot have $n = 1$ and $k = 14$, nor can we have $n = 14$ and $k = 1$. However it is also impossible to have $n = 2$ and $k = 7$ because for $n = 2$ the "no more than two" restriction would forbid an entry of 5.
Thus we have $n = 7$ and $k = 2$.

7. Suppose we define a Bloom filter with $m = 4$ and $k = 3$. We insert $n = 2$ keys.

   What is the probability that the resulting bit array has `1`s in exactly two indices and `0`s elsewhere?

   Solution:
   We're making 6 index choices in $\{0, 1, 2, 3\}$ and need to get exactly two indices.
   Overall there are 4^6 choices each with probability $1/4^6$.
   For the ones of interest there are $C(4, 2) = 6$ possibilities:
   We get 6 index choices in $\{0, 1\}$ for which there are $2^6 = 64$ choices but not all 0 or all 1, thus $64 - 2 = 62$ choices.
   The same for $\{0, 2\}$, $\{0, 3\}$, $\{1, 2\}$, $\{1, 3\}$, and $\{2, 3\}$ for a total of $6(62) = 372$ choices.
   This yields an overall probability of $372(1/4^6)$.

8. Suppose we define a very small Bloom filter with $m = 4$ and $k = 2$.

   (a) We insert $n = 2$ keys. What is the probability that the resulting bit array is `1000`? Show your work.

   Solution:
   When we insert $n = 2$ keys with $k = 2$ hash functions we are essentially picking four indices from $\{0, 1, 2, 3\}$ with repeats permitted and setting those positions to `1`.
   To obtain `1000` we must choose index 0 four times in order.
   The number of ways to choose four 0s in order is 1 and it has probability $(1/4)^4 = 1/256$.

   (b) We insert $n = 2$ keys. What is the probability that the resulting bit array is `1111`? Show your work.

   Solution:
   When we insert $n = 2$ keys with $k = 2$ hash functions we are essentially picking four indices from $\{0, 1, 2, 3\}$ with repeats permitted and setting those positions to `1`.
   To obtain `1111` we must choose each index exactly once.
   The number of ways to choose each index exactly once is $4! = 24$.
   Each has probability $1/256$ and they are disjoint so the result is $24/256 = 3/32$.

   (c) We insert $n = 2$ keys. What is the probability that the resulting bit array is `1010`? Show your work.

   Solution:
   When we insert $n = 2$ keys with $k = 2$ hash functions we are essentially picking four indices from $\{0, 1, 2, 3\}$ with repeats permitted and setting those positions to `1`.
   To obtain `1010` we must choose from indices $\{0, 2\}$ four times in order but the result must include at least one 0 and one 2.
   The number of ways to choose from indices $\{0, 2\}$ four times in order is $2^4 = 16$ but we have to exclude $0, 0, 0, 0$ and $2, 2, 2, 2$ so there are 14 ways.
   Each has probability $1/256$ and they are disjoint so the result is $14/256 = 7/128$.

9. Suppose we define a Bloom filter with $m = 3$ and $k = 2$. We insert $n = 2$ keys.

   What is the probability that the resulting bit array is `111`?

   Solution:
   We're making $nk = 4$ index choices from $\{0, 1, 2\}$ and need to get every single index at least once.

   Overall there are $3^4 = 81$ ordered choices of 3 indices each with probability $1/81$, for example `0`, `2`, `0`, `1` and `1`, `1`, `1`, `0`, etc.

   We want to get at least one each of `0`, `1`, and `2`. Thus we don't want any of the following:

   - All `0`s and/or `1`s, for which there are $2^4 = 16$ choices.
   - All `0`s and/or `2`s, for which there are $2^4 = 16$ choices.
   - All `1`s and/or `2`s, for which there are $2^4 = 16$ choices.

   This would be $3(16) = 48$ but this counts all `0`s, all `1`s, and all `2`s so it's only 45.
   Thus there are $81 - 45 = 36$ choices in which we get at least one each of `0`, `1`, and `2`.

   This yields an overall probability of $36/81 = 4/9$.

10. Suppose a Bloom Filter has table size $m = 1000$ and we use $k = 50$ hash functions. What is the maximum number of keys we can store in it to ensure that the probability of a false positive is at most 0.9? Show your calculation and only approximate at the final step.

    Use the incorrect (but simple and approximate) formula for this.

    Solution:
    we need to ensure that:

$$
\begin{aligned}
\left(1 - \left(1 - \frac{1}{1000}\right)^{50n}\right)^{50} &\leq 0.9 \\
1 - (0.999)^{50n} &\leq \sqrt[50]{0.9} \\
(0.999)^{50n} &\geq 1 - \sqrt[50]{0.9} \\
(0.999)^{n} &\geq \sqrt[50]{1 - \sqrt[50]{0.9}} \\
n &\leq \log_{0.999}\left(\sqrt[50]{1 - \sqrt[50]{0.9}}\right) \\
n &\leq 123.207
\end{aligned}
$$

    Since $n \in \mathcal{Z}$ we must have $n \leq 123$.

11. Consider the incorrect (but simple and approximate) formula for a false positive. For fixed $m = 100$ and $n = 50$ plot this as a function of k for $0 \leq k \leq 20$. Comment on why its shape makes intuitive sense.

Solution:
We have:

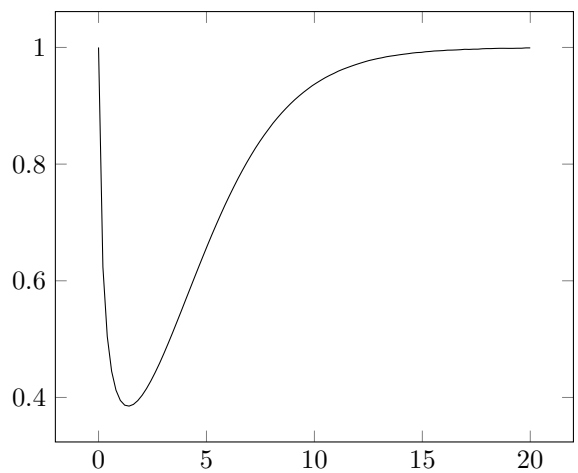

Figure 1: Plot of False Positive Probability

This makes sense because:

- For $k = 0$ no bits would be set for any keys and no bits would be checked for any keys, meaning we would always get false positives, vacuously.
- For very small k because each key sets very few bits there is a very small chance of a false positive because, for any other key there is a smaller chance of a match.
- For very large k because each key sets so many bits there is a larger chance of a false positive because, for any other key, so many bits will already be set.

12. Suppose we define a very small Bloom filter with unknown $m = 3$ and $k = 2$. We insert $n = 2$ keys. What is the probability that the resulting bit array is `011`? Show your work.

Solution:
When we insert $n = 2$ keys with $k = 2$ hash functions we are essentially picking 4 indices between 0 and 2 with repeats permitted and setting those positions to `1`.
To obtain `011` we must choose between indices 1 and 2 a total of 4 times in order but the result must include at least one 1 and one 2.
The number of ways to choose four 0s and 2s in order is $2^4 = 16$ but we have to exclude $1, 1, 1, 1$ and $2, 2, 2, 2$ so there are 14 ways.
Each has probability $1/3^4 = 1/81$ and they are disjoint so the result is $14/81$.